

FaceVista

AI-Powered Facial Surgery Visualization

& Consultation Platform

● Computer Vision

● Generative AI

● LangChain + LangGraph

DOCUMENT TYPE

Project Documentation

VERSION

1.0 · September 2026

LEAD ENGINEER

Talha Habib

AI ENGINEER

Abdul Ghafoor

FaceVista is a visualization and communication tool only. It does not constitute medical advice.

0. Table of Contents

1. Abstract

2. Introduction

- 2.1 Project Overview
- 2.2 Problem Statement
- 2.3 Proposed Solution
- 2.4 Project Objectives
- 2.5 Project Scope

3. Literature Review and Existing Systems

- 3.1 Existing Systems Overview
- 3.2 Comparative Analysis
- 3.3 Research Gap

4. System Analysis

- 4.1 Target Users
- 4.2 Functional Requirements
- 4.3 Non-Functional Requirements
- 4.4 Use Case Diagrams
- 4.5 Supported Facial Areas

5. System Design

- 5.1 High-Level System Architecture
- 5.2 Technology Stack
- 5.3 Image Processing Pipeline
- 5.4 AI and Machine Learning Pipeline
- 5.5 Database Schema
- 5.6 API Design
- 5.7 Frontend Design
- 5.8 Surgeon Dashboard

6. Implementation Details

- 6.1 Computer Vision Component
- 6.2 AI Image Generation
- 6.3 LangChain Integration
- 6.4 LangGraph Workflow Orchestration
- 6.5 MCP Integration
- 6.6 n8n Automation Workflows
- 6.7 Object Storage
- 6.8 AI Consultation Report Generation
- 6.9 Safety Layer

7. Testing Strategy

8. Risk Analysis and Mitigation

9. Ethical Considerations

10. Development Roadmap

11. Conclusion

12. References

01 Abstract

FaceVista is an AI-powered facial surgery visualization and consultation platform designed to assist patients and cosmetic surgeons in exploring possible facial modifications through personalized visual simulations. The system employs Computer Vision techniques, including facial landmark detection and facial region segmentation, to analyze uploaded facial photographs. Users select a specific facial region — beginning with the nose in Version 1.0 — and the system generates multiple visual simulations representing subtle, moderate, and pronounced modifications.

The AI workflow is orchestrated through LangChain and LangGraph, while the Model Context Protocol (MCP) provides modular tool integration. A Stable Diffusion-based generative model produces identity-preserving simulations. A dedicated surgeon dashboard allows medical professionals to review generated simulations and provide clinical feedback. Workflow automation through n8n manages consultation scheduling, notifications, and reporting.

■ **IMPORTANT: FaceVista is designed strictly as a visualization and communication tool. It shall not independently diagnose conditions, recommend surgery, or guarantee surgical outcomes. All medical decisions shall remain the sole responsibility of qualified professionals.**

02 Introduction

2.1 Project Overview

FaceVista allows a user to upload standardized facial photographs including front-facing and profile images. Computer Vision techniques detect important facial landmarks and analyze the structure of different facial regions.

Variation	Description
Original Face	The unmodified uploaded photograph
Subtle	A minimal, conservative change
Moderate	A balanced, mid-range change
Pronounced	A more significant visual change

2.2 Problem Statement

Patients considering facial surgery frequently encounter difficulty articulating the specific appearance they desire to a surgeon. Existing facial-editing applications are primarily designed for entertainment rather than a structured patient-surgeon consultation workflow. These tools typically lack:

- Standardized facial analysis based on anatomical landmarks
- Multiple graduated variations for systematic comparison
- Structured consultation support connecting patient preferences to professional review
- Identity-preserving modifications that maintain the patient's authentic appearance
- Workflow automation integrating simulation, reporting, and appointment management

2.3 Proposed Solution

The proposed system acts as an AI-assisted facial visualization platform. The general workflow proceeds from patient image upload through computer vision analysis, facial landmark detection, facial region identification, AI simulation engine, multiple possible visualizations, patient comparison interface, AI-generated consultation summary, surgeon review dashboard, and final professional consultation.

Critical Design Principle: The AI shall not make any medical decision. The system helps patients and surgeons communicate and visualize possibilities, while the qualified surgeon remains solely responsible for medical assessment and clinical decisions.

2.4 Project Objectives

#	Objective
O1	Analyze facial images using Computer Vision techniques
O2	Detect and map important facial landmarks with high accuracy
O3	Identify and segment different facial regions from detected landmarks
O4	Allow users to select a specific facial area for visualization

#	Objective
O5	Generate multiple realistic visual simulations at varying intensities
O6	Preserve the patient's identity while modifying the selected region
O7	Provide a side-by-side comparison interface for original and simulated appearances
O8	Generate structured consultation summaries for professional review
O9	Provide a dedicated surgeon dashboard for simulation review and feedback
O10	Automate consultation workflow including scheduling, notifications, and reporting
O11	Maintain an explicit distinction between visualization and medical advice

2.5 Project Scope

In Scope	Out of Scope
Facial image upload and quality validation	Medical diagnosis or surgical recommendation
Facial landmark detection and segmentation	Real-time video-based simulation
Nose region visualization (v1.0 MVP)	3D facial modeling (future versions)
Multiple simulation generation	EHR integration
Identity-preserving AI image generation	Mobile native applications (web-based only)
Side-by-side comparison interface	
AI-generated consultation reports	
Surgeon review dashboard	
Workflow automation	

03 Literature Review & Existing Systems

3.1 Existing Systems Overview

Crisalix

Attribute	Detail
Technology	3D/4D mesh simulation
Target Users	Board-certified surgeons
Strengths	High-fidelity 3D modeling, immersive patient experience
Limitations	High cost, proprietary, no AI-agentic workflow

Canfield Vectra

Attribute	Detail
Technology	Proprietary 3D imaging hardware
Target Users	High-volume surgical practices
Strengths	Exceptional precision, clinical-grade analysis
Limitations	Very high cost, requires in-clinic capture

FaceTouchUp

Attribute	Detail
Technology	AI-based 2D photo morphing
Target Users	Surgeons, clinics, individual patients
Strengths	Highly accessible, affordable, quick turnaround
Limitations	2D only, no structured workflow, no multi-variation

AEDIT

Attribute	Detail
Technology	AI facial measurement + 3D try-on
Target Users	Patients & practitioners
Strengths	Medically informed analysis, individualized reports
Limitations	Limited surgeon-side tools, no agentic AI workflow

FaceApp

Attribute	Detail
Technology	Deep learning neural transfer
Target Users	General consumers

Attribute	Detail
Strengths	Polished output, instant results, large user base
Limitations	No clinical relevance, entertainment-focused only

3.2 Comparative Analysis

Feature	Crisalix	Vectra	FaceTouch Up	AEDIT	FaceApp	FaceVista
Simulation Type	3D/4D	3D	2D	3D	2D	2D + AI
Multiple Variations	Limited	Limited	No	Limited	No	Yes (3 levels)
AI Orchestration	No	No	No	No	No	LangChain + LangGraph
Landmark Analysis	Partial	Yes	No	Yes	No	MediaPipe (468pts)
Identity Preservation	Moderate	High	Low	Moderate	Low	High (IP-Adapter)
Surgeon Dashboard	Basic	Yes	No	No	No	Yes (dedicated)
Consult. Workflow	Manual	Manual	No	Partial	No	Automated (n8n)
Modular Architecture	No	No	No	No	No	Yes (MCP)
Open Source	No	No	No	No	No	Yes
Cost	High	Very High	Low	Free/Premium	Free	Self-hosted

3.3 Research Gap

- Gap 1: No existing system employs an AI agent framework (LangChain/LangGraph) for end-to-end workflow.
- Gap 2: Existing systems produce single simulations; FaceVista generates multiple graduated variations.
- Gap 3: Existing systems are monolithic; FaceVista adopts modular architecture using MCP.
- Gap 4: Consultation lifecycle is managed manually; FaceVista automates through n8n.
- Gap 5: Clinical-grade solutions are proprietary; FaceVista provides an open-source alternative.

04 System Analysis

4.1 Target Users

User Type	Description
Patients (Primary)	Individuals exploring possible facial surgery. Upload photos, select regions, review simulations, initiate consultation requests.
Surgeons (Professional)	Licensed surgeons using the system to understand patient expectations. Access dedicated dashboard for simulation review.
Cosmetic Clinics (Secondary)	Clinics integrating the platform into consultation and patient-management workflows for intake, communication, and appointments.

4.2 Functional Requirements

ID	Requirement	Priority
FR01	Allow users to upload facial images in JPEG and PNG formats	High
FR02	Validate uploaded images for quality, resolution, and facial positioning	High
FR03	Detect and map facial landmarks using Computer Vision	High
FR04	Segment and identify distinct facial regions from detected landmarks	High
FR05	Allow users to select a specific facial region for simulation	High
FR06	Generate three simulation variations (subtle, moderate, pronounced)	High
FR07	Preserve the patient's identity across all simulations	High
FR08	Display original and simulated images in a side-by-side comparison	High
FR09	Generate an AI-powered consultation summary report	Medium
FR10	Provide a surgeon dashboard for reviewing simulations	Medium
FR11	Allow surgeons to add comments and feedback to simulations	Medium
FR12	Support consultation appointment scheduling	Medium
FR13	Send automated email notifications for key workflow events	Low
FR14	Provide user authentication and role-based access control	High
FR15	Support conversational interaction through an AI assistant	Medium

4.3 Non-Functional Requirements

ID	Requirement	Category
NFR-01	Process and return simulation results within 60 seconds	Performance
NFR-02	Support at least 50 concurrent users	Scalability
NFR-03	Maintain 99.5% uptime availability	Availability

ID	Requirement	Category
NFR-04	All facial images shall be encrypted at rest and in transit	Security
NFR-05	Implement role-based access control (RBAC)	Security
NFR-06	Accessible via modern web browsers (Chrome, Firefox, Safari, Edge)	Compatibility
NFR-07	Provide responsive design for desktop and tablet devices	Usability
NFR-08	Log all access events for audit purposes	Security
NFR-09	All generated simulations include a visualization-only disclaimer	Compliance
NFR-10	Support horizontal scaling of the AI processing pipeline	Scalability

4.5 Supported Facial Areas

Version 1.0 implements the **Nose** region exclusively as the MVP. Subsequent versions introduce: Chin & Jaw (v2.0), Cheeks & Lips (v3.0), Eyes & Eyebrows (v4.0).

Nose Region — Version 1.0 MVP

Parameter	Description
Nose Width	Adjustment of overall nasal width
Nose Bridge	Modification of dorsal bridge profile
Nose Tip	Refinement of nasal tip shape and projection
Dorsal Hump	Reduction or enhancement of dorsal hump
Nose Projection	Adjustment of forward nasal projection
Overall Proportion	Modification of nose-to-face ratio

05 System Design

5.1 High-Level System Architecture

User / Surgeon (Browser)
Next.js + React Frontend
FastAPI Backend · Redis Cache
LangGraph Workflow Engine
LangChain Orchestration Safety Layer MCP Tool Layer
Computer Vision Module Generative AI Pipeline n8n Automation
Landmark Detection Facial Segmentation Stable Diffusion + Inpainting
Cloudflare R2 Object Storage · PostgreSQL Database

5.2 Technology Stack

Layer	Technology	Purpose
AI / Agent	LangChain	AI application orchestration and LLM integration
AI / Agent	LangGraph	Stateful multi-step workflow orchestration
AI / Agent	MCP	Modular tool and service integration
Computer Vision	OpenCV	Image processing, alignment, cropping, masking, blending
Computer Vision	MediaPipe Face Mesh	Facial landmark detection (468 points)
Computer Vision	InsightFace	Face detection, analysis, identity preservation
Generative AI	Stable Diffusion	Base generative model for facial simulation
Generative AI	ControlNet	Structural guidance for controlled image generation
Generative AI	IP-Adapter	Identity-preserving adapter for consistent generation
Backend	Python 3.11 + FastAPI	Asynchronous REST API framework
Database	PostgreSQL + SQLAlchemy	Relational database with ORM
Storage	Cloudflare R2	S3-compatible object storage (zero egress fees)
Caching	Redis	Session management, task queuing, result caching
Automation	n8n	Business workflow automation (email, scheduling)
Frontend	Next.js 14 + Tailwind CSS	React-based SSR with utility CSS
Infrastructure	Docker + Docker Compose	Containerization and multi-container orchestration

5.4 AI and Machine Learning Pipeline

Stage 1 — Pretrained Models

Model	Purpose
MediaPipe Face Mesh	Facial landmark detection
Stable Diffusion v1.5/XL	Base generative model
ControlNet (face model)	Structural guidance during generation
IP-Adapter FaceID	Identity preservation during modification

Stage 2 — Fine-Tuning

Task	Approach
Nose modification	Fine-tune inpainting model on nose-modification datasets
Identity preservation	Fine-tune IP-Adapter for facial surgery context
Quality assessment	Train classifier for simulation quality scoring

5.5 Database Schema Overview

Entity	Key Fields	Relationships
USER	id, email, role, is_active	Creates Consultations, Uploads Images
SURGEON_PROFILE	user_id, license_number, specialization	Linked to USER
CONSULTATION	patient_id, surgeon_id, selected_area, status	Has Images, Simulations, Reports
FACIAL_IMAGE	user_id, consultation_id, storage_url, landmarks_data	Source for Simulations
SIMULATION	consultation_id, variation_level, identity_score, quality_score	Receives Feedback
SIMULATION_FEEDBACK	simulation_id, surgeon_id, comment, rating	Linked to SIMULATION
APPOINTMENT	consultation_id, patient_id, surgeon_id, scheduled_at	Linked to CONSULTATION
CONSULTATION_REPORT	consultation_id, summary, pdf_url	Generated from CONSULTATION
AUDIT_LOG	user_id, action, resource_type, ip_address	Tracks all USER actions

5.6 API Design

Method	Endpoint	Description
POST	/api/v1/auth/register	Register a new user account
POST	/api/v1/auth/login	Authenticate and receive JWT token
POST	/api/v1/images/upload	Upload a facial photograph
POST	/api/v1/images/{id}/analyze	Trigger facial analysis on an image

Method	Endpoint	Description
POST	/api/v1/simulations/generate	Generate simulation variations
GET	/api/v1/simulations/{id}	Retrieve a specific simulation
POST	/api/v1/simulations/{id}/feedback	Submit surgeon feedback
POST	/api/v1/consultations	Create a new consultation
POST	/api/v1/consultations/{id}/report	Generate consultation report
POST	/api/v1/appointments	Schedule an appointment
POST	/api/v1/assistant/chat	Conversational AI assistant interaction
WS	/api/v1/assistant/ws	Real-time assistant communication

06 Implementation Details

6.1 Computer Vision Component

Technology	Role	Justification
MediaPipe Face Mesh	Facial landmark detection	468 3D landmarks in real-time; lightweight; well-documented
OpenCV	Image processing	Industry standard for manipulation, alignment, masking, blending
InsightFace	Face analysis & identity	Robust face detection; identity embedding for preservation scoring

Key Facial Landmark Groups

Landmark Group	Point Range	Usage
Nose bridge	Points 6, 197, 195, 5	Bridge profile analysis
Nose tip	Points 1, 2, 98, 327	Tip shape and projection
Nose wings	Points 129, 358, 49, 279	Nose width measurement
Left eye	Points 33, 133, 160, 144	Reference alignment
Right eye	Points 362, 263, 387, 373	Reference alignment
Mouth	Points 61, 291, 0, 17	Lower face reference
Chin	Points 152, 377, 148	Face proportion baseline
Face oval	Points 10, 338, 297, 332, 284	Face boundary detection

6.2 AI Image Generation

Variation	Modification Intensity	Denoising Strength	ControlNet Weight
Subtle	15–25% of computed range	0.3 – 0.4	0.8
Moderate	40–60% of computed range	0.5 – 0.6	0.6
Pronounced	75–90% of computed range	0.7 – 0.8	0.4

Identity Preservation Strategy

- IP-Adapter FaceID: Extracts a facial identity embedding and conditions the generative model.
- Tight Masking: Only the selected facial region is modified; surrounding areas preserved.
- Feathered Blending: Boundary between modified and original regions blended with soft gradient.
- Identity Similarity Scoring: Cosine similarity computed by InsightFace. Threshold: 0.85.

6.3 LangChain Tool Integration

Tool Name	Function	Input	Output
analyze_face	Run CV analysis on image	Image ID	Landmarks, measurements, quality score

Tool Name	Function	Input	Output
generate_simulation	Generate facial simulation	Image ID, region, level	Simulation image URL
create_report	Generate consultation report	Consultation ID	Report PDF URL
check_availability	Query surgeon availability	Surgeon ID, date range	Available slots
schedule_appointment	Book a consultation	Patient ID, surgeon ID, datetime	Appointment confirmation

6.5 MCP Tool Registry

MCP Tool	Input Schema	Output Schema	Description
facial_analysis_tool	{image_id: str}	{landmarks, measurements}	Complete facial analysis
image_quality_tool	{image_id: str}	{score, issues}	Image quality assessment
landmark_detection_tool	{image_id, region}	{points: list}	Region-specific landmarks
simulation_tool	{image_id, region, level}	{simulation_url, scores}	Generate simulation
patient_report_tool	{consultation_id}	{report_url}	Generate report
appointment_tool	{patient_id, surgeon_id, datetime}	{appointment_id, status}	Schedule appointment

6.6 n8n Automation Workflows

Workflow	Trigger	Steps
Post-Simulation Notification	Simulation Complete (Webhook)	Generate PDF → Store R2 → Email Patient → Log Event
Appointment Scheduling	Consultation Request	Check Availability → Create Appointment → Confirm → Notify Surgeon
Consultation Reminder	Scheduled (24h Before)	Query Upcoming → Send Reminder (Patient) → Send Reminder (Surgeon)
Surgeon Alert	New Consultation (Webhook)	Identify Surgeon → Send Email → Dashboard Push Notification

6.7 Object Storage

Policy	Value	Rationale
Image Encryption	AES-256 at rest	Patient data sensitivity
Access Method	Pre-signed URLs (15-min expiry)	Prevent unauthorized access
Temporary Files	Auto-delete after 24 hours	Minimize data retention
Backup	Daily incremental to secondary region	Disaster recovery

6.9 Safety Layer

Category	Prohibited Output
Diagnosis	"You need surgery" or any diagnostic statement
Guarantee	"This surgery will definitely give you this result"
Judgment	"Your face is abnormal" or any derogatory assessment
Recommendation	"You should get rhinoplasty" or any medical recommendation
Comparison	"Your nose is worse than average" or comparative judgments

Required Disclaimer on all outputs: "This is a computer-generated visual simulation for consultation and communication purposes only. It does not represent a prediction or guarantee of actual surgical outcomes. All medical decisions must be made by a qualified, licensed medical professional."

07 Testing Strategy

7.1 Unit Testing

Component	Test Focus	Framework
Facial Landmark Detection	Accuracy against annotated ground truth	pytest
Image Quality Assessment	Correct classification of acceptable/unacceptable images	pytest
Region Mask Generation	Correct mask boundaries for nose region	pytest
Identity Score Calculation	Accurate cosine similarity computation	pytest
API Endpoint Handlers	Request validation, response formatting, error handling	pytest + httpx
Database Models	CRUD operations, constraint validation	pytest + SQLAlchemy
Safety Filter	Detection and removal of prohibited content	pytest

7.3 System Testing Scenarios

Test Scenario	Description	Expected Outcome
Complete Patient Journey	Upload → Analysis → Simulation → Report → Consultation Request	All stages complete successfully
Invalid Image Handling	Upload a non-face image	System rejects with appropriate error
Low Quality Image	Upload blurred or poorly lit image	Quality feedback and re-upload request
Surgeon Review Flow	Surgeon views patient, reviews simulations, adds comments	Data correctly displayed and persisted
Appointment Scheduling	Patient requests, surgeon confirms	Appointment created with notifications

7.5 Performance Targets

Metric	Target	Tool
Image Upload Latency	< 3 seconds for 5 MB image	Locust / k6
Facial Analysis Time	< 5 seconds per image	Custom benchmark
Simulation Generation (3 vars)	< 60 seconds	Custom benchmark
API Response Time (p95)	< 500 ms for non-AI endpoints	Locust / k6
Concurrent User Support	50 simultaneous users	Locust / k6
Database Query Time (p95)	< 100 ms	pgbench
Frontend Load Time	< 3 seconds (LCP)	Lighthouse

08 Risk Analysis & Mitigation

#	Risk	Prob.	Impact	Mitigation
R1	Unrealistic simulation outputs	High	High	Quality scoring + auto-rejection; ControlNet structural guidance; identity scoring
R2	Identity loss during modification	Medium	Critical	Identity threshold >0.85; IP-Adapter FaceID; tight masking
R3	Patient psychological harm	Medium	High	Mandatory disclaimers; safety layer; clear "illustrative only" messaging
R4	Demographic bias in simulations	High	High	Test across Fitzpatrick I–VI, ages, genders, ethnicities; curate diverse training data
R5	Data breach / Privacy violation	Low	Critical	Encrypt at rest/transit; pre-signed URLs; RBAC; audit logging; auto-deletion
R6	High computational cost	High	Medium	Redis caching; batch processing; model quantization; async job management
R7	Misuse as medical advice	Medium	Critical	Safety layer; mandatory disclaimers; no diagnostic language; terms of service
R8	Model availability dependency	Medium	Medium	Abstract model interface; fallback options; version-pin dependencies
R9	Scalability bottleneck	Medium	Medium	Queue-based processing; horizontal GPU scaling; async jobs
R10	Integration complexity	High	Medium	Modular architecture; integration testing; Docker-based deployment

09 Ethical Considerations

9.1 Patient Autonomy & Informed Consent

The system obtains explicit informed consent before processing any facial image. Patients are clearly informed that images will be processed by AI, simulations are illustrative only, they retain the right to request deletion of all data at any time, and no medical diagnosis is being provided.

9.2 Non-Maleficence

The system avoids causing harm: no language suggesting a patient "needs" surgery; no negative characterization of natural appearance; no comparison against idealized beauty standards; simulations represent plausible modifications, not unattainable ideals.

9.3 Fairness & Bias

The system is evaluated for equitable performance across skin tones (Fitzpatrick Types I–VI), ages (young adults to older individuals), gender presentations, facial structures, and ethnic backgrounds. Performance disparities are documented transparently and addressed as priority improvements.

9.4 Transparency

Users are informed that AI generates the simulations. The system does not present AI-generated content as medical imaging. Limitations are clearly communicated at every stage.

9.5 Data Minimization

Processing intermediates (masks, landmarks) are deleted after simulation completion. Patients may delete all data upon request. Retention periods are clearly communicated and enforced.

9.6 Professional Responsibility

All simulations are marked as requiring professional review. The surgeon dashboard facilitates — not replaces — clinical judgment. All medical decisions are explicitly deferred to licensed practitioners.

10 Development Roadmap

Phase 1 — Proof of Concept (Month 1–2)

Deliverable	Description
Image upload API	FastAPI endpoint for facial image upload
Face detection	OpenCV/MediaPipe face detection integration
Landmark detection	MediaPipe Face Mesh integration with 468-point extraction
Nose landmark extraction	Isolate nose-specific landmarks from full face mesh
Basic frontend	Next.js page with upload capability and landmark visualization
Database setup	PostgreSQL schema for users and images
Docker configuration	Containerized development environment

Phase 2 — Visualization Engine (Month 3–4)

Deliverable	Description
Nose segmentation	Generate precise nose region masks from landmarks
Modification parameters	Compute nose-specific modification ranges
Stable Diffusion integration	Inpainting pipeline for nose region modification
ControlNet integration	Structural guidance for anatomically consistent generation
Three-variation generation	Subtle, moderate, and pronounced simulation outputs
Identity preservation	IP-Adapter integration with similarity scoring
Comparison interface	Side-by-side simulation comparison UI
Object storage	Cloudflare R2 integration for image storage

Phase 3 — AI Agent Integration (Month 5–6)

Deliverable	Description
LangChain agent	Conversational AI assistant for guided interaction
LangGraph workflow	Stateful multi-step workflow from upload to report
Tool integration	Connect CV and simulation tools to LangChain agent
State management	Full workflow state tracking and persistence
Error recovery	Automatic retry and graceful failure handling

Phase 4 — MCP & Tool Architecture (Month 7–8)

Deliverable	Description
MCP server	MCP protocol implementation for tool exposure
facial_analysis_tool	MCP-wrapped facial analysis capability
simulation_tool	MCP-wrapped simulation generation
patient_report_tool	MCP-wrapped consultation report generation
appointment_tool	MCP-wrapped appointment scheduling

Phase 5 — Clinic Workflow (Month 9–10) · Phase 6 — Surgeon Dashboard (Month 11–12) · Phase 7 — Model Improvement (Month 13+)

11 Conclusion

FaceVista presents a comprehensive AI-powered facial surgery visualization and consultation platform that addresses a genuine communication gap between patients and cosmetic surgeons. By combining Computer Vision, generative AI, agentic workflow orchestration, and clinic automation, the system provides a structured, multi-variation visualization experience that empowers informed consultation.

The platform distinguishes itself through AI agent orchestration (LangChain + LangGraph), modular tool architecture (MCP), automated clinic workflow (n8n), and systematic multi-variation generation. The explicit safety layer and ethical design principles ensure responsible operation within the healthcare-adjacent domain.

■ **IMPORTANT: FaceVista is designed exclusively as a visualization and communication tool. It shall never independently diagnose conditions, recommend surgery, or guarantee surgical outcomes. All medical decisions shall remain the sole responsibility of qualified, licensed medical professionals.**

12 References

- [1] Lugaresi, C., Tang, J., Nash, H., et al. (2019). "MediaPipe: A Framework for Building Perception Pipelines." arXiv:1906.08172.
- [2] Rombach, R., Blattmann, A., Lorenz, D., et al. (2022). "High-Resolution Image Synthesis with Latent Diffusion Models." CVPR.
- [3] Zhang, L., Rao, A., & Agrawala, M. (2023). "Adding Conditional Control to Text-to-Image Diffusion Models." ICCV.
- [4] Deng, J., Guo, J., Xue, N., & Zafeiriou, S. (2019). "ArcFace: Additive Angular Margin Loss for Deep Face Recognition." CVPR.
- [5] Ye, H., Zhang, J., Liu, S., et al. (2023). "IP-Adapter: Text Compatible Image Prompt Adapter." arXiv:2308.06721.
- [6] Bradski, G. (2000). "The OpenCV Library." Dr. Dobb's Journal of Software Tools.
- [7] Guo, Y., Zhang, L., et al. (2016). "MS-Celeb-1M: A Dataset and Benchmark for Large-Scale Face Recognition." ECCV.
- [8] Harrison Chase et al. (2022). "LangChain: Building applications with LLMs through composability." GitHub.
- [9] Tiangolo, S. (2018). "FastAPI: Modern, fast web framework for building APIs with Python." GitHub.
- [10] n8n GmbH. (2019). "n8n: Workflow Automation Tool." <https://n8n.io>.